

Activity, Intent & Activity Navigation in Android

1. What is an Activity?

An Activity is a component of an Android application that represents a screen with which the user can interact.

Activity = One screen of an Android application.

For ex: a student application can have:

LoginActivity → Login Screen

HomeActivity → Home Screen

ProfileActivity → Profile Screen

SettingsActivity → Settings Screen

Real-Life Example

Think of an application like a book

- Application = Complete book
- Activity = One page/screen
- Button = Action to move somewhere
- Intent = Instruction to open another page

Example

A Login screen can be represented by:

LoginActivity



activity_login.xml

The Java file contains the logic, while the XML file defines the UI.

LoginActivity.java

+

activity_login.xml



Login Screen

Activity represents a screen of an Android application.

2. What is Activity Navigation?

Activity Navigation is the process of moving from one Activity/screen to another Activity in an Android application.

Activity Navigation = Moving from one app screen to another.

Example

Suppose your application has:

Login Screen



[Login]



Home Screen

When the user clicks the Login button, the application opens the Home Activity.

Another Example

Home Screen



[My Profile]



Profile Screen

Real-Life Example

Think about a shopping application:

Home



Products



Product Details



Cart



Payment

Every movement from one screen to another is navigation.

Activity Navigation means changing/moving between different screens of an Android application.

3. What is an Intent?

An Intent is a messaging object used by Android to request an action from another component, such as opening another Activity.

Intent is used to tell Android that we want to perform an action, such as opening another Activity.

Intent as a message/instruction.

Button Click



Create Intent



Specify Next Activity



`startActivity()`



Next Screen Opens

Q. example: Open Profile Screen on Button Click

HOME SCREEN



[View Profile]



Intent



PROFILE SCREEN

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
```

```
    xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:orientation="vertical"
```

```
    android:gravity="center"
```

```
    android:padding="20dp">
```

```
    <TextView
```

```
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
```

```
        android:text="Student App"
```

```
        android:textSize="28sp"
```

```
        android:textStyle="bold" />
```

```
<Button
    android:id="@+id/profileButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:text="View Profile" />
</LinearLayout>
```

MainActivity.java

```
package com.example.myapplication;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
```

```
    Button profileButton;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        profileButton = findViewById(R.id.profileButton);
        profileButton.setOnClickListener(v -> {
```

```
        Intent intent = new Intent(
            MainActivity.this,
            ProfileActivity.class
        );
        startActivity(intent);
    });
}
```

Create Second Activity

Right-click java package → New → Activity → Empty Views Activity

Name it:

ProfileActivity

Android Studio will create:

ProfileActivity.java

activity_profile.xml

activity_profile.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
```

```
    xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
android:orientation="vertical"
```

```
android:gravity="center"
```

```
android:padding="20dp">
```

```
<TextView
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:text="My Profile"
```

```
    android:textSize="28sp"
```

```
    android:textStyle="bold" />
```

```
<TextView
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:layout_marginTop="20dp"
```

```
    android:text="Name: Sneha"
```

```
    android:textSize="18sp" />
```

```
<TextView
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:text="Course: B.Tech AI & DS"
```

```
    android:textSize="18sp" />
```

```
</LinearLayout>
```

ProfileActivity.java

```
package com.example.myapplication;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

public class ProfileActivity extends AppCompatActivity {

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_profile);

    }

}
```

Understand the Important Java Code

Step 1 — Create Intent

```
Intent intent = new Intent(

    MainActivity.this,

    ProfileActivity.class );
```

This tells Android:

“I am currently in MainActivity and I want to open ProfileActivity.”

Step 2 — Start Activity

```
startActivity(intent);
```

This tells Android:

“Open ProfileActivity now.”